

v1.0Higgs Boson Deep Dive Emulation

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Higgs Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Higgs Boson Deep Dive Emulation V2.0.md

Revision Necessity — Four Drivers

- Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
- New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
- Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
- Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Higgs self-coupling	Not measured	ATLAS 2024: HH→bbyy 3.4σ evidence. Lambda=0.48±0.28. V2.0: V(H)=-μ²H²+λH⁴ force law added	HIGH	First measurement of Higgs potential shape is major missing feature
H→Zγ decay	Not discovered in V1.0	ATLAS+CMS 2023: H→Zγ at 3.4σ (BR=3.4×10^-3). New decay mode vertex added	HIGH	Confirmed Higgs decay absent from V1.0 simulation
Higgs mass	m_H = 125.25 GeV (2022)	m_H = 125.20±0.11 GeV (ATLAS+CMS 2024 combined)	LOW	Sub-percent precision update, minor parameter adjustment
Top Yukawa	y_t ~ 1 (estimate)	ATLAS+CMS tTH 2023: y_t = 1.04±0.09 — 9% precision. Updates gg→H loop calculation	MEDIUM	Higgs production via top loop uses this coupling

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** $H\bar{Z}\gamma$, Higgs
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Higgs Boson Deep Dive Emulation V1.0 archived | V2.0 is the active specification as of June 2026